La Scuola d'Italia / Dr. Huson / IB Math
22 April 2026

First & Last name:

5.15 Classwork: Logarithms (no calculator)

1. [Maximum mark: 12]

   (a) Calculate the value of each of the following logarithms:

   (i) $\log_4 \frac{1}{64}$

   (ii) $\log_{25} 5$

   (iii) $\log_2 \sqrt{32}$ [6]

   (b) Let $c = \log_3 m$, where $m > 0$. Write down each of the following expressions in terms of c.

   (i) $\log_3 m^2$

   (ii) $\log_3 27m$

   (iii) $\log_9 m$ [6]

2. [Maximum mark: 7]

Solve $\log_2(x-3) = \log_4(x+9)$.

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Answer all questions on lined paper. Please start each question on a new page.

3. [Maximum mark: 15]

   Consider the series $\ln x + p\ln x + \dfrac{1}{3}\ln x + \ldots$, where $x \in \mathbb{R}$, $x > 1$ and $p \in \mathbb{R}$, $p \neq 0$.

   (a) Consider the case where the series is geometric.

   (i) Show that $p = \pm\dfrac{1}{\sqrt{3}}$.

   (ii) Given that $p > 0$ and $S_\infty = 3 + \sqrt{3}$, find the value of x. [5]

   (b) Now consider the case where the series is arithmetic with common difference d.

   (i) Show that $p = \dfrac{2}{3}$.

   (ii) Write down d in the form $k\ln x$, where $k \in \mathbb{R}$.

   (iii) The sum of the first n terms of the series is $-3\ln x$. Find the value of n. [10]

4. [Maximum mark: 14]

   Let $g(x) = p^x + q$, for x, p, $q \in \mathbb{R}$, $p > 1$. The point $A(0, a)$ lies on the graph of g.

   Let $f(x) = g^{-1}(x)$. The point B lies on the graph of f and is the reflection of point A in the line $y = x$.

   (a) Write down the coordinates of B. [2]

   The line L_1 is tangent to the graph of f at B.

   (b) Given that $f'(a) = \dfrac{1}{\ln p}$, find the equation of L_1 in terms of x, p and q. [5]

   The line L_2 is tangent to the graph of g at A and has equation $y = (\ln p)\,x + q + 1$.

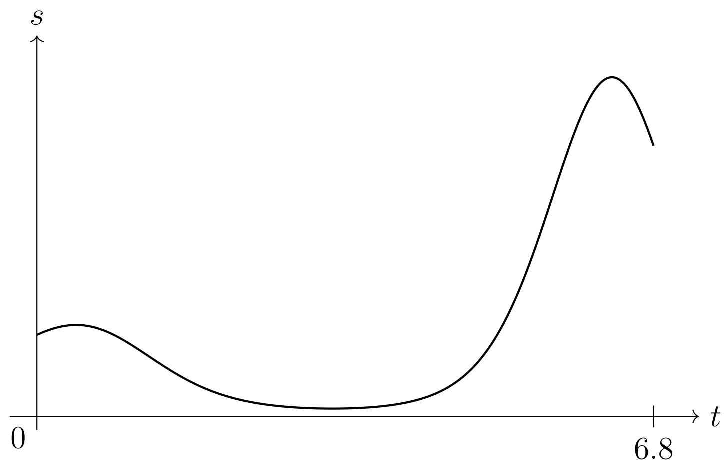
   The line L_2 passes through the point $(-2, -2)$.

   The gradient of the normal to g at A is $\dfrac{1}{\ln \frac{1}{3}}$.

   (c) Find the equation of L_1 in terms of x. [7]

5. [Maximum mark: 16]

A particle moves in a straight line. Its displacement, s metres, from a fixed point P at time t seconds is given by $s(t) = 3(t + 2)^{\cos t}$, for $0 \leq t \leq 6.8$, as shown in the following graph.



(a) Find the particle's initial displacement from the point P. [2]

(b) Find the particle's velocity when $t = 2$. [2]

(c) Determine the intervals of time when the particle is moving away from the point P. [5]

The acceleration of the particle is zero when $t = b$ and $t = c$, where $b < c$.

(d) Find the value of b and the value of c. [4]

(e) Find the total distance travelled by the particle for $b \leq t \leq c$. [3]